

CS481 HW1

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March 6, 2026

Q1

$(1017)_{10} == (11\ 1111\ 1001)_2$

A:

64 bytes means we need $\log_2 64$ bits for the Page Offset, so 6, so the first 6 are ignored. With the given TLB we assume the address is parsed

TT HOO OOOO

Tag = $(3)_{10} = (11)_2$

Index = $(3)_{10} = (11)_2$

B:

TLB MISS (NO set = 3)

C:

VPN is the Upper Bits

VPN = $(15)_{10} = (1111)_2$

D:

PFN = $(14)_{10}$ from Page Table

E:

Physical Address = $\text{PFN} \cdot 64 + \text{offset} = 14 \cdot 64 + 57 = (953)_{10} = (11\ 1011\ 1001)_2$

16 Line cache implies 4 Tag Bits (MSB's), 4 Index Bits (Middle Bits), 2 Offset Bits (LSB's), TT TTH HOO

Tag = $(14)_{10} = (1110)_2$

Index = $(14)_{10} = (1110)_2$

Offset = $(1)_{10} = (1)_2$ **F:**

In the Cache, We Can see IDX 14. Tag 14, Offset 1 = 17

Q2

$(1186)_{10} == (100\ 1010\ 0010)_2$

A:

Same Setup from Before, the parsed address is TTT HOO OOOO

Tag = $(4)_{10} = (100)_2$

Index = $(2)_{10} = (10)_2$

B:

TLB MISS (Set 2, Tag 4 is empty)

C:

VPN is upper bits

$$\text{VPN} = (18)_{10} = (1\ 0010)_2$$

D:

$$\text{PFN} = (13)_{10}$$

E:

$$\text{PA} = \text{PFN} \cdot 64 + \text{offset} = 13 \cdot 64 + 34 = (866)_{10} = (11\ 0100\ 0010)_2$$

Same cache Set-Up, TT TTII IIOO

$$\text{Tag} = (13)_{10} = (1101)_2$$

$$\text{Index} = (8)_{10} = (1000)_2$$

$$\text{Offset} = (2)_{10} = (10)_2$$

F:

In Cache, we see that the value is 27

Q3

$$(34)_{10} == (10\ 0010)_2$$

A:

The Tag and Index are both 0 since the address only contains the Offset bits.

B:

TLB MISS (Set 0, Tag 0 has an Invalid Bit)

C:

VPN has a Valid Bit in the Page Table

D:

$$\text{PFN} = 0$$

E:

$$\text{PA} = \text{PFN} \cdot 64 + \text{offset} = 0 \cdot 64 + 34 = (34)_{10} = (10\ 0010)_2$$

Set up is Same, TT TTII IIOO

$$\text{Tag} = 0$$

$$\text{Index} = (8)_{10} = (1000)_2$$

$$\text{Offset} = (2)_{10} = (10)_2$$

F:

Index 8 has tag 13, CACHE MISS